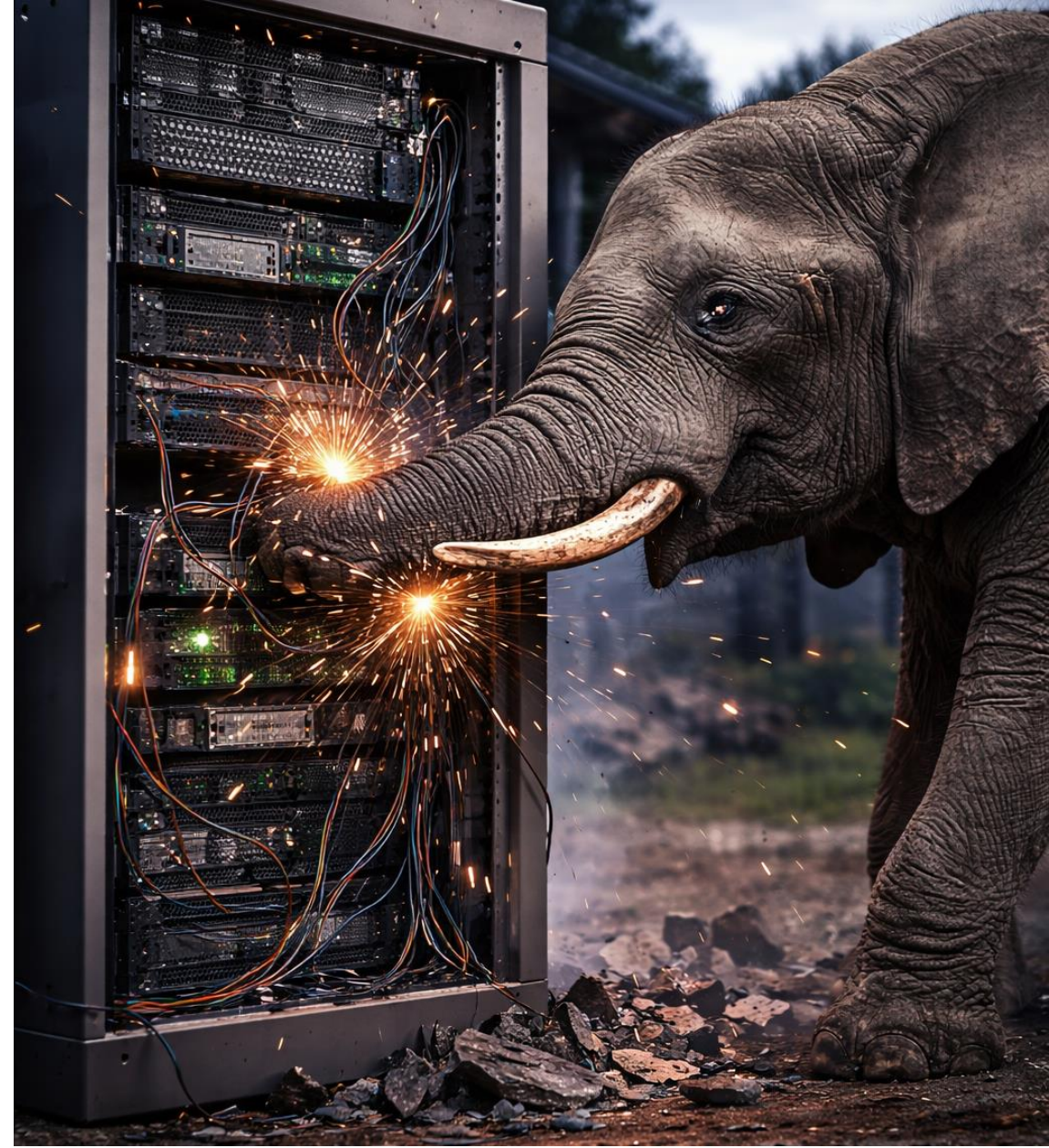


Platform Ownership Endorsements (POE)

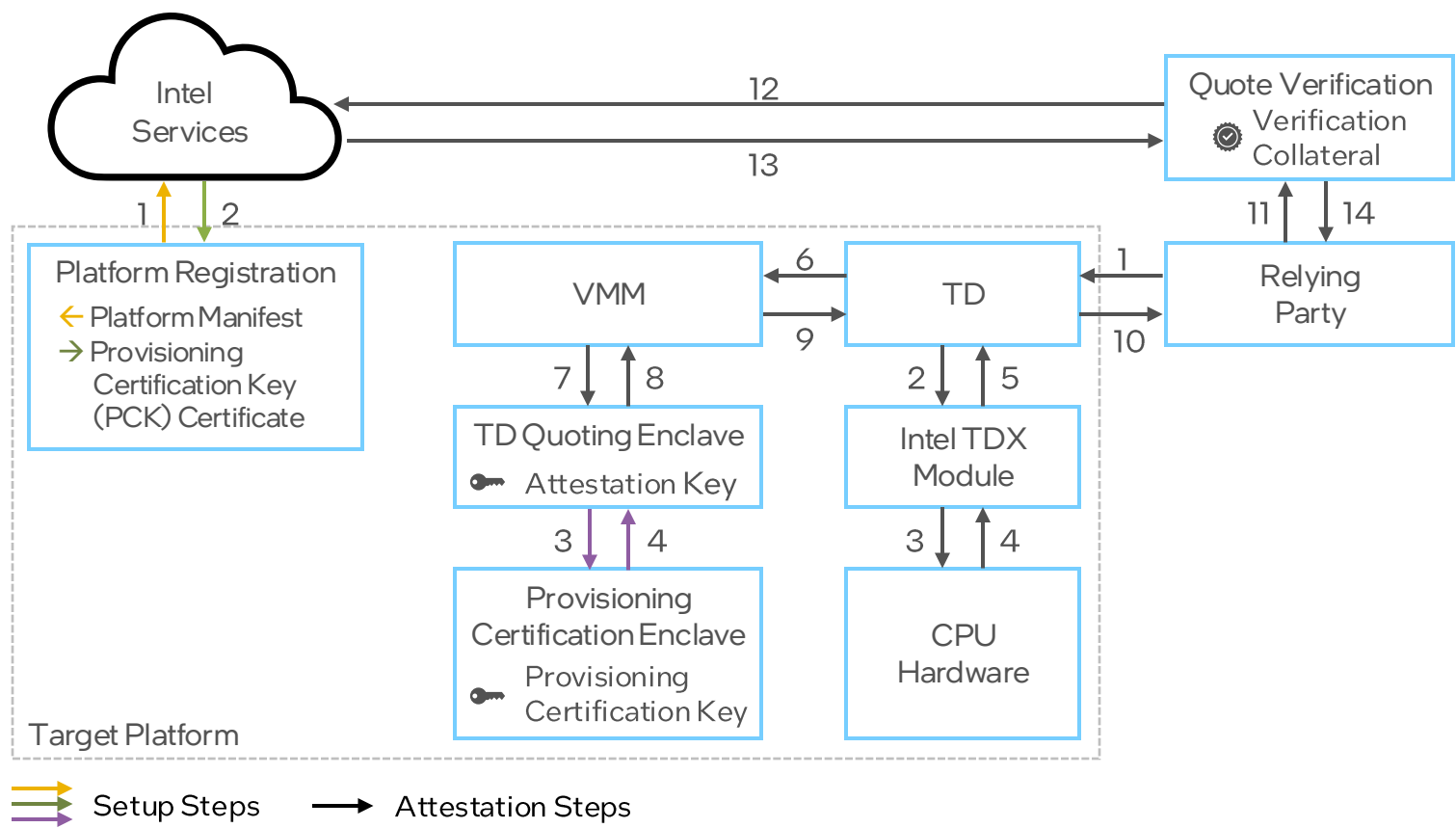
Physical Attacks on Intel® SGX/TDX – the Elephant in the Room

- Physical access changes the threat model
 - Memory probing & bus observation
 - Fault injection
 - Cold-boot
 - Ciphertext observation & tampering
- Customers
 - Might trust CSP to not launch physical attacks
 - Currently do not have a way to verify physical ownership of a platform
- Goal of Platform Ownership Endorsements (POE): Link attestation evidence to a trusted physical owner

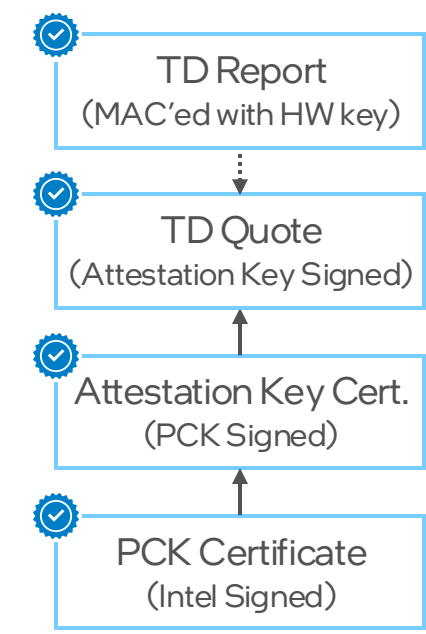


Remote Attestation (Intel TDX)

Process:

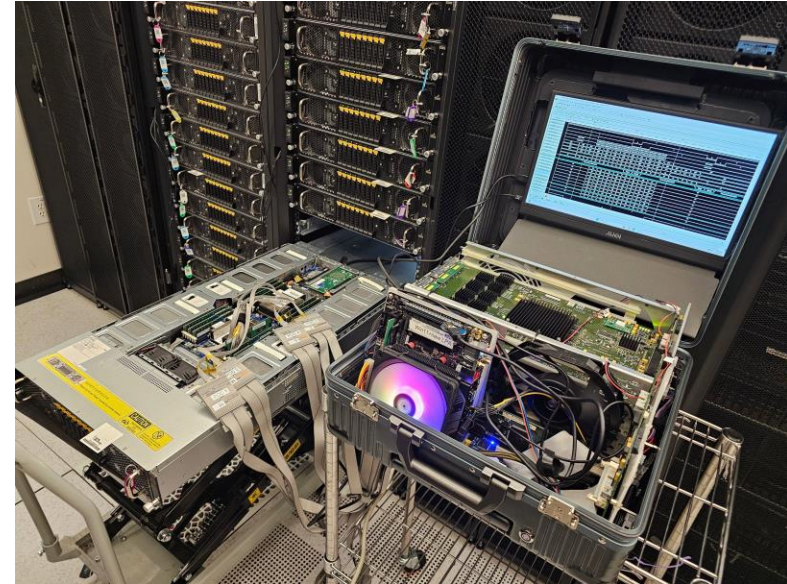


Certificate Chain:



Physical Attack on Intel SGX/TDX

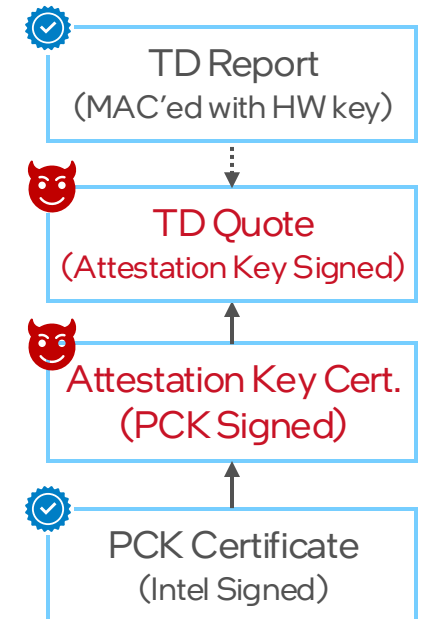
- Approach: Physically deploy DRAM Bus interposer to observe memory transactions



- Background: Intel SGX and Intel TDX use standardized, deterministic AES-XTS encryption, i.e., the same plaintext data encrypted to the same cache line is encrypted to the same ciphertext
- Leakage: With such a physical attack, researchers were able to leak the PCK private key

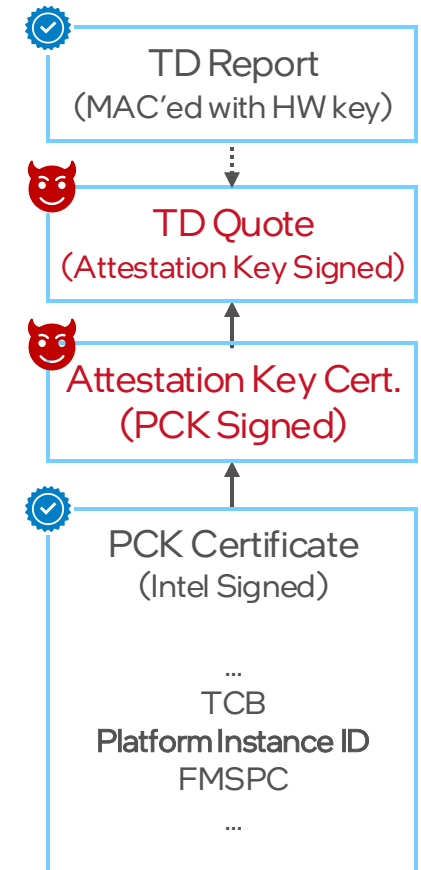
Consequence of a Leaked Attestation Key

- Attacker is able to generate TD Quotes with arbitrary content, e.g.,
 - Arbitrary REPORTDATA (user defined data often used for PK hash)
 - Arbitrary TDATTRIBUTES (includes debug mode bit)
 - Arbitrary MRTD (measurement of TD virtual firmware)
- Such a TD Quote is considered valid by a Relying Party because the boot chain is still rooted in Intel
- Note, once the attestation key is identified as compromised, Intel can revoke the platform and individual CPUs of the platform
 - Before revocation, there is no way to recognize a spoofed TD Quote

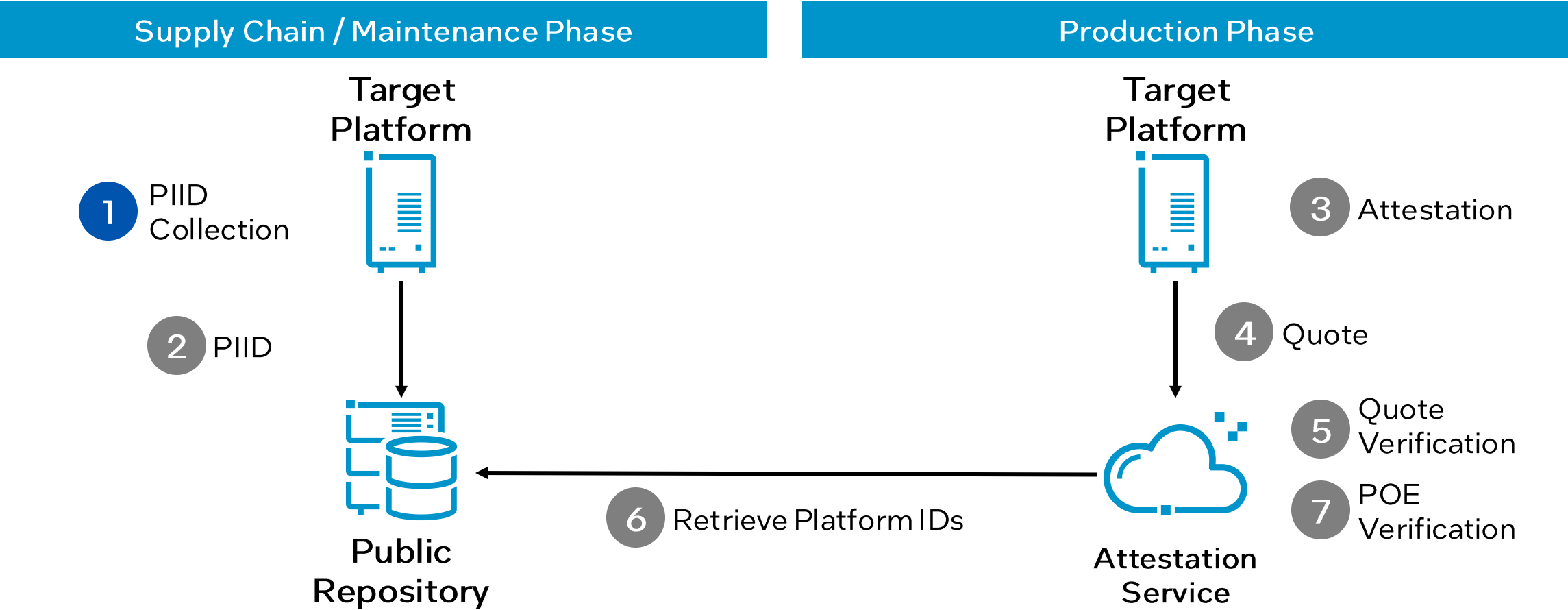


Hardening Remote Attestation

- Relying Party must
 - be able to recognize the physical owner of the machine
 - trust physical owner to prevent physical attacks
- Key insights for solution:
 - Attacker cannot modify PCK Certificate
 - PCK Certificate contains unique Platform Instance ID (PIID)
 - PIID is generated randomly on first boot
 - PIID survives reboots
 - PIID can be reset by explicitly performing an “Intel SGX Factory Reset”
 - PIID is contained in the Platform Manifest, PCK Certs, and TD quotes

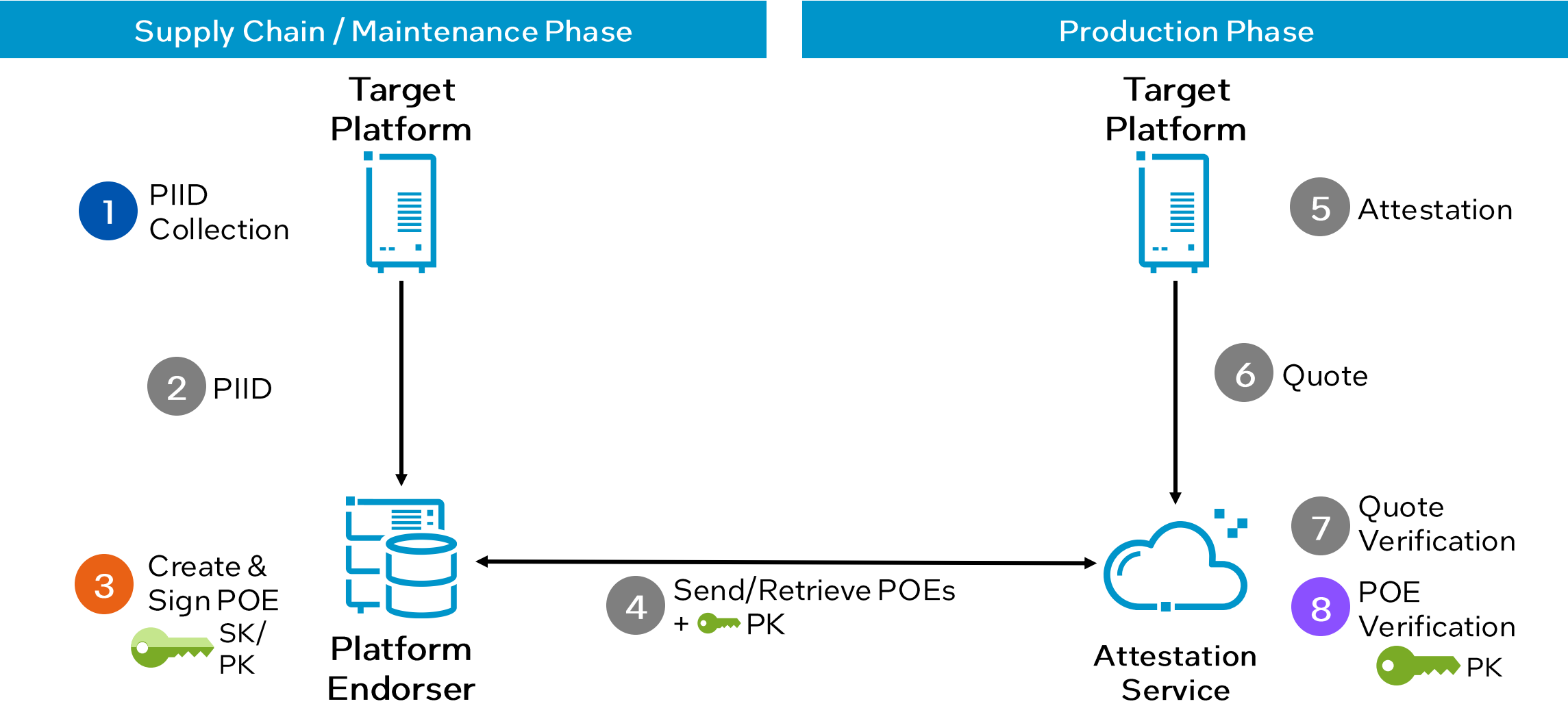


POE Flow – Strawman

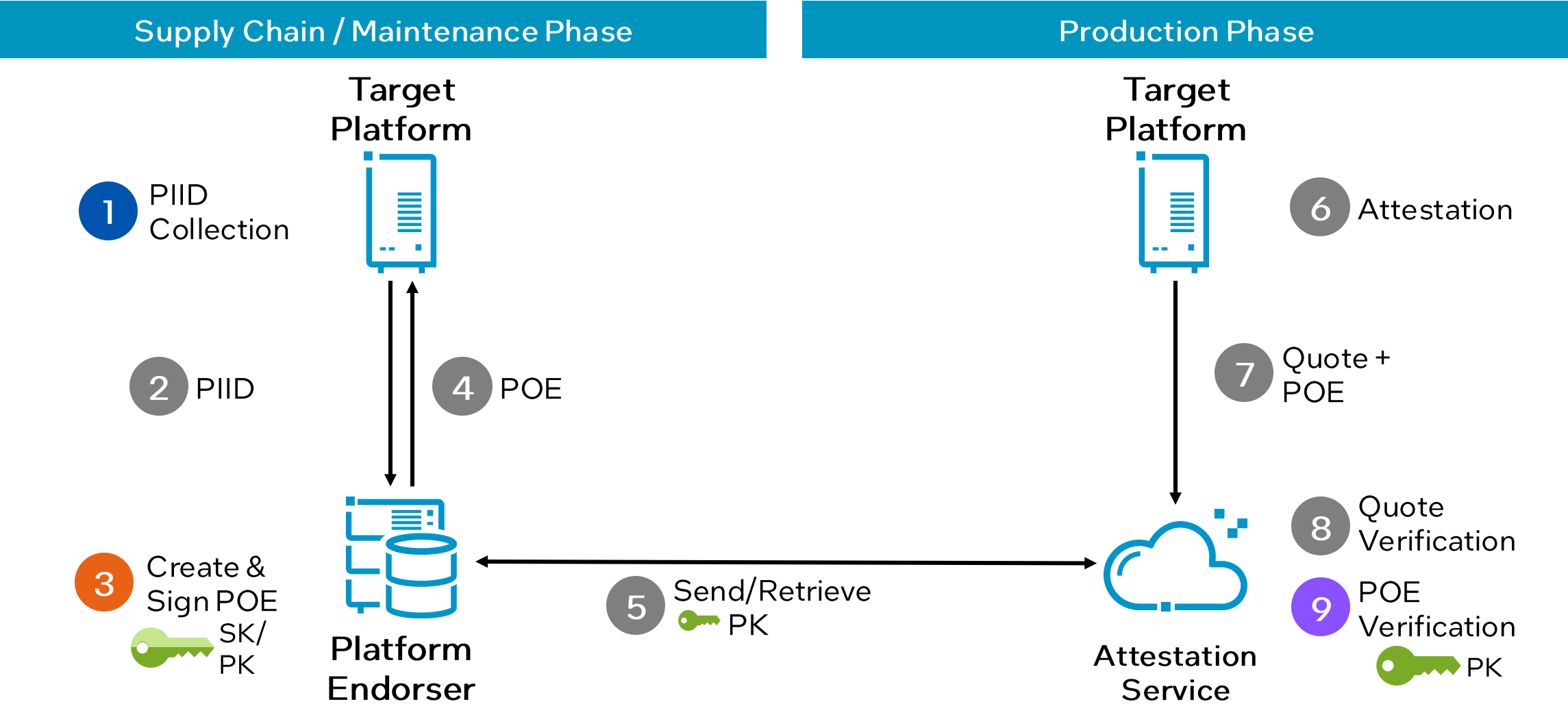


Issue: PIIDs are not integrity protected and CSP might not be willing to publish IDs

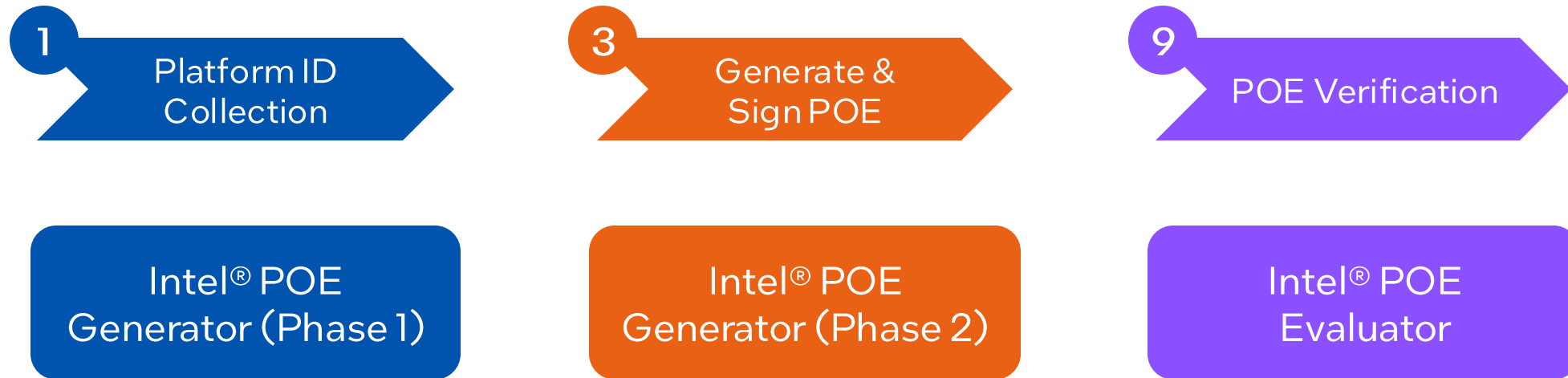
POE Flow – Short Term



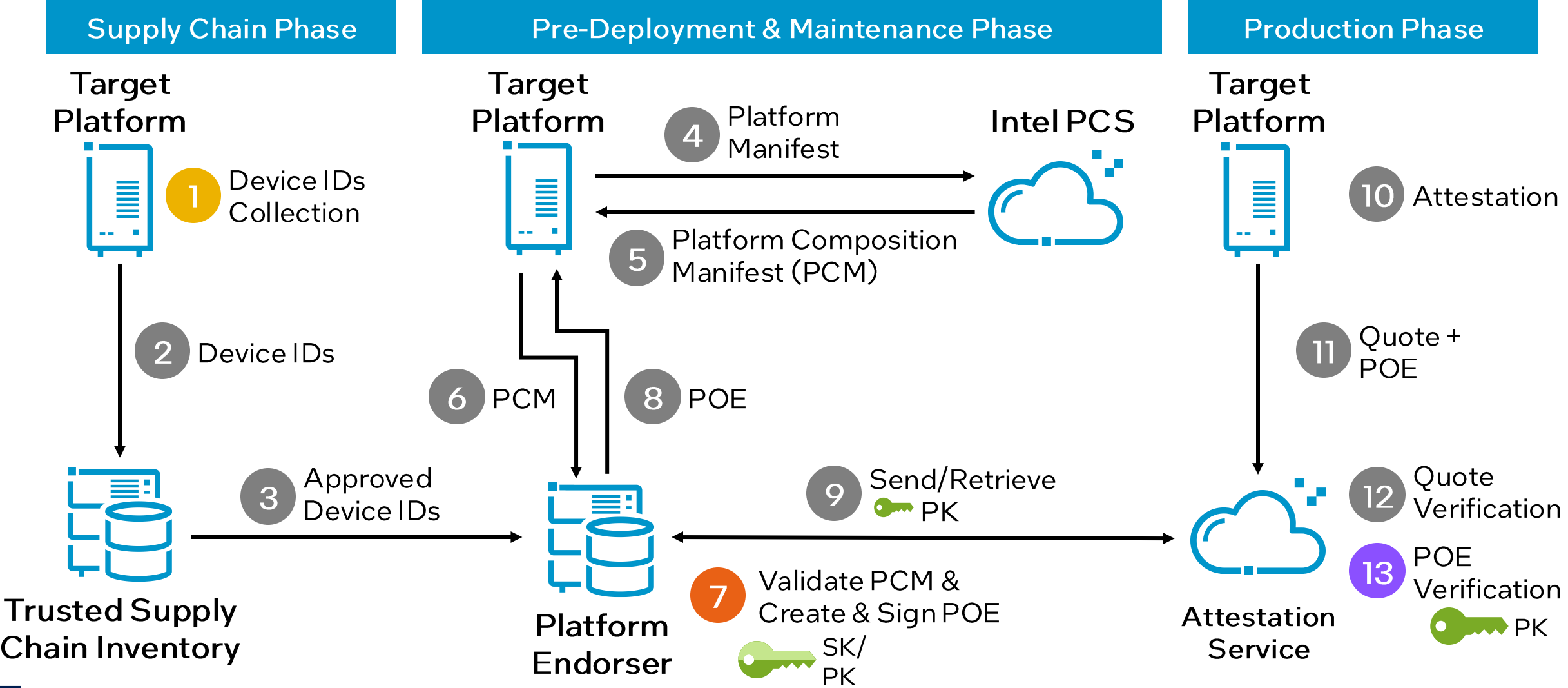
POE Flow – Longer Term



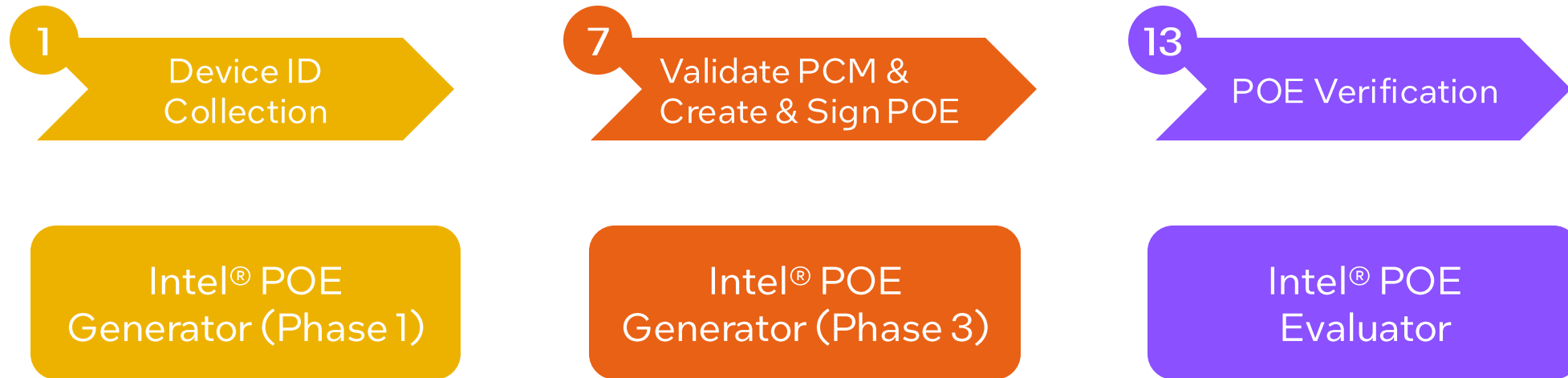
Intel Tools for POE Deployment



POE Flow with Platform Composition Endorsements



Intel Tools for POE Deployment with Platform Composition Endorsements



Intel POE Specification

The screenshot shows the Intel website's 'Platform Ownership Endorsements for Confidential Computing' page. The page has a blue header with the Intel logo and navigation links. Below the header, there's a breadcrumb trail: 'Developers > Topics & Technologies > Software Security Guidance > Feature Documentation > Platform Ownership Endorsements'. The main content area has a blue banner with the title 'Platform Ownership Endorsements for Confidential Computing'. Below the banner, there's a table with one row of metadata: ID (871281), Updated (2/2/2026), Version (1.0), and Public. To the left of the main content, there's a sidebar with a table of contents including 'Terminology', 'Introduction', 'Platform Ownership Endorsement for Intel® Software Guard Extensions (Intel® SGX) and Intel® Trust Domain Extensions (Intel® TDX)', 'POE Creation with Platform Composition Endorsements', 'Summary', 'Resources', and 'Footnotes'. The 'Terminology' section is expanded, showing a table with two rows: 'CoRIM' defined as 'Concise Reference Integrity Manifest' and 'Intel® SGX / Intel® TDX DCAP' defined as 'Intel® Software Guard Extensions Data Center Attestation Primitives / Intel® Trust Domain Extensions for Data Center Attestation Primitives'. Below the table of contents, there's a 'Key Takeaways' section with a bullet point: 'Platform Ownership Endorsement (POE) enables remote parties to establish who is...'. To the right of the 'Terminology' table, there are icons for printing and emailing the document.

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Platform Ownership Endorsements for Confidential Computing

ID	Updated	Version	
871281	2/2/2026	1.0	Public

Terminology Introduction

> Platform Ownership Endorsement for Intel® Software Guard Extensions (Intel® SGX) and Intel® Trust Domain Extensions (Intel® TDX)

POE Creation with Platform Composition Endorsements

Summary Resources Footnotes

Key Takeaways

- Platform Ownership Endorsement (POE) enables remote parties to establish who is...

Terminology

Term	Definition
CoRIM	Concise Reference Integrity Manifest
Intel® SGX / Intel® TDX DCAP	Intel® Software Guard Extensions Data Center Attestation Primitives / Intel® Trust Domain Extensions for Data Center Attestation Primitives

- [Platform Ownership Endorsements for Confidential Computing](#)



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